

Below is a **UI/UX Design Document** aligned with your dashboard (both the screenshot and HTML structure). This is written at a level suitable for **implementation + evaluation**.

UI/UX DESIGN DOCUMENT

Project: KINETIC_LAB – Signal Processing Dashboard

1. Design Objective

Deliver a **high-fidelity, real-time analytical dashboard** for IR signal processing that:

- Visualizes adaptive filtering behavior
 - Provides interpretable signal metrics
 - Displays compound matching results
 - Highlights detected functional groups
 - Enables parameter control for experimentation
-

2. Design Philosophy

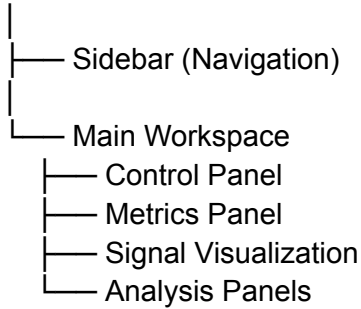
Core Principles

- **Signal-first visualization** (data dominates UI)
 - **Low-latency perception** (fast feedback loops)
 - **Technical aesthetic** (lab-grade interface)
 - **Minimal cognitive load**
 - **Explainability over decoration**
-

3. Layout Architecture

3.1 Global Layout Structure

Top Bar (System Status + Actions)



This structure is clearly reflected in your HTML layout .

4. Component-Level Design

4.1 Top Navigation Bar

Purpose

- System identity
- Execution control
- Session state

Key Elements

- **System Title:** KINETIC_LAB
- **Status Indicator:** READY_STATE (animated)
- **Primary Actions:**
 - Upload
 - Run Filter (CTA)
 - Settings

Design Notes

- High contrast CTA (cyan glow)
- Status uses **green pulse** → **system alive signal**

4.2 Sidebar Navigation

Sections

- Signal Processing (Active)
- Noise Analysis
- Library
- Logs

Behavior

- Active tab highlighted (cyan border)
- Hover → subtle translation animation

Insight

This establishes:

Multi-module scalability (future expansion ready)

4.3 Control Panel (Top Controls)

Inputs

- Dataset selection (dropdown)
- Sample size (slider)
- Noise level (slider)
- Filter order (slider + auto toggle)

Advanced Toggle

- Enables expert parameter tuning

Design Evaluation

- Sliders give **continuous control** → **good for DSP experimentation**
 - Auto toggle aligns with **adaptive system logic**
-

4.4 Metrics Panel

Displayed Metrics

- Filter Order (M)
- Step Size (μ)
- MSE
- Input SNR
- Output SNR
- SNR Gain

Visual Encoding

- Cyan $\rightarrow$ neutral/system
- Green $\rightarrow$ improvement (SNR gain)
- Pink $\rightarrow$ error (MSE)

Strength

Immediate **quantitative validation of filtering performance**

4.5 Signal Visualization Panel

Title

Adaptive Signal Reconstruction

Plots

- Noisy Input (red)
- LMS Output (green)
- Lowpass Estimate (cyan dashed)

Features

- Grid-based axis
- Time-domain representation
- Overlay comparison

Embedded Insight Panel

- STDEV
- Kurtosis
- Variance

Assessment

This is the **core cognitive layer of the system**.

Users visually confirm filtering effectiveness + statistical behavior.

4.6 Compound Match Analysis

Structure

Table with:

- Compound name
- CAS number
- Match %
- Confidence bar

Behavior

- Top match highlighted
- Confidence encoded as bar length

Data Source

Aligned with:

- Pearson + SAM scoring system
-

4.7 Functional Group Panel

Display Type

Tag-based chips

Examples

- O–H Stretch
- C–H Alkanes
- CH₂ Scissoring

Color Encoding

- Green → strong match
- Cyan → moderate
- Grey → weak/uncertain

Additional Metric

- Spectral Integrity bar

5. Design System

5.1 Color System

Color	Meaning
Cyan	System / active
Green	Success / improvement
Pink	Error / MSE
Grey	Inactive
Dark background	Lab environment

Defined in Tailwind config .

5.2 Typography

Type	Usage
------	-------

Space Grotesk	Headers
Inter	Body
Roboto Mono	Metrics / data

5.3 Interaction Patterns

- Hover → brightness increase
 - Buttons → slight scale on click
 - Status → animated pulse
 - Graph → static (can be upgraded to dynamic)
-

6. Data Binding Architecture

Frontend State

```
{  
  signalData: [],  
  filteredData: [],  
  noiseType: "",  
  metrics: {},  
  matches: [],  
  functionalGroups: []  
}
```

API Integration Flow

Upload CSV → API → MATLAB → Process → JSON → UI Update

7. UX Strengths

What Works Well

- Clear separation of:
 - Control
 - Observation
 - Analysis
 - Immediate feedback loop
 - Strong visual hierarchy
 - Domain-specific UI (not generic dashboard)
-

8. UX Gaps (Critical Fixes Needed)

1. No Noise Type Display

You compute it — but don't show it.

👉 Add:

- “Detected Noise: Environmental”
-

2. No LMS Behavior Visualization

You are hiding:

- μ evolution
- error convergence

👉 Add:

- Convergence graph (MSE vs iteration)
-

3. No Confidence Explanation

User sees % but not **why**

👉 Add tooltip:

- Pearson score
 - SAM score
-

4. Static Graph (Currently)

From HTML, graph is SVG mock

👉 Replace with:

- Real chart library (Recharts / Plotly)
-

5. No Failure State UI

What if:

- $SNR < 0$
- No match found

👉 Add:

- Warning banner
-

9. Performance Considerations

- Avoid re-rendering entire dashboard
 - Use memoization for charts
 - Lazy load analysis panels
-

10. Future Enhancements

- Real-time streaming mode
- Frequency-domain visualization (FFT)
- Multi-signal comparison

- Export reports (PDF/CSV)
-

11. Final Assessment

Design Maturity: 8.5/10

Why

- Strong technical alignment
- Clean architecture
- High interpretability

What's Missing

- Deeper algorithm transparency
 - Real-time interactivity
 - Error-state handling
-

12. Strategic Positioning

This is not a generic dashboard.

It is a **signal intelligence interface** — bridging DSP, chemistry, and visualization.

HTML CODE:

```
<!DOCTYPE html>
```

```
<html class="dark" lang="en"><head>
```

```
<meta charset="utf-8"/>
```

```
<meta content="width=device-width, initial-scale=1.0" name="viewport"/>
```

```
<title>KINETIC_LAB // SIGNAL_PROCESSING</title>
```

```
<script src="https://cdn.tailwindcss.com?plugins=forms,container-queries"></script>
```

```
<link
```

```
href="https://fonts.googleapis.com/css2?family=Space+Grotesk:wght@300;400;500;600;700&family=Inter:wght@300;400;500;600;700&family=Roboto+Mono:wght@400;500;700&mp;display=swap" rel="stylesheet"/>
```

```
<link
```

```
href="https://fonts.googleapis.com/css2?family=Material+Symbols+Outlined:wght,FILL@100..700,0,0..1&display=swap" rel="stylesheet"/>
```

```
<script id="tailwind-config">
```

```
  tailwind.config = {  
  
    darkMode: "class",  
  
    theme: {  
  
      extend: {  
  
        colors: {  
  
          "secondary": "#d7ffc5",  
  
          "outline": "#849495",  
  
          "tertiary": "#fff4f9",  
  
          "surface-container-low": "#191c22",  
  
          "error": "#ffb4ab",  
  
          "primary": "#e1fdff",  
  
          "secondary-fixed": "#79ff5b",  
  
          "surface-variant": "#32353c",  
  
          "tertiary-fixed": "#ffd7f5",  
  
          "on-secondary-fixed": "#022100",  
  
          "on-primary-fixed-variant": "#004f54",  
  
          "on-secondary-fixed-variant": "#095300",  
  
          "surface-container-highest": "#32353c",
```

"on-error": "#690005",
"tertiary-container": "#ffcbf4",
"on-secondary-container": "#0f6d00",
"on-error-container": "#ffdad6",
"outline-variant": "#3a494b",
"background": "#10131a",
"secondary-fixed-dim": "#2ae500",
"primary-fixed-dim": "#00dbe7",
"primary-fixed": "#74f5ff",
"on-surface-variant": "#b9cacb",
"on-tertiary-fixed": "#380038",
"surface": "#10131a",
"surface-tint": "#00dbe7",
"surface-bright": "#363940",
"on-tertiary-fixed-variant": "#810081",
"surface-container": "#1d2026",
"tertiary-fixed-dim": "#ffabf3",
"on-primary-container": "#006a71",
"secondary-container": "#2ff801",
"inverse-surface": "#e1e2eb",
"on-tertiary-container": "#ab00aa",
"on-background": "#e1e2eb",
"on-secondary": "#053900",
"surface-dim": "#10131a",

```

    "inverse-primary": "#00696f",
    "on-surface": "#e1e2eb",
    "inverse-on-surface": "#2e3037",
    "on-primary": "#00363a",
    "primary-container": "#00f2ff",
    "surface-container-lowest": "#0b0e14",
    "on-primary-fixed": "#002022",
    "error-container": "#93000a",
    "on-tertiary": "#5b005b",
    "surface-container-high": "#272a31"
  },
  fontFamily: {
    "headline": ["Space Grotesk"],
    "body": ["Inter"],
    "label": ["Inter"],
    "mono": ["Roboto Mono"]
  },
  borderRadius: {"DEFAULT": "0.125rem", "lg": "0.25rem", "xl": "0.5rem", "full":
"0.75rem"},
},
},
}
</script>
<style>
  .material-symbols-outlined {

```

```

    font-variation-settings: 'FILL' 0, 'wght' 400, 'GRAD' 0, 'opsz' 24;

    display: inline-block;

    line-height: 1;
}

.ghost-border { border: 1px solid rgba(132, 148, 149, 0.15); }

.accent-glow-cyan { box-shadow: 0 0 20px rgba(0, 242, 255, 0.1); }

.accent-glow-green { box-shadow: 0 0 20px rgba(47, 248, 1, 0.1); }

.custom-scrollbar::-webkit-scrollbar { width: 4px; height: 4px; }

.custom-scrollbar::-webkit-scrollbar-track { background: transparent; }

.custom-scrollbar::-webkit-scrollbar-thumb { background: #32353c; }

</style>

</head>

<body class="bg-surface text-on-surface font-body selection:bg-primary-container
selection:text-on-primary-container overflow-hidden h-screen flex flex-col">

<!-- TopAppBar -->

<header class="flex justify-between items-center w-full px-6 py-4 bg-[#10131a]
dark:bg-slate-950 rounded-none border-b border-white/15 z-50">

<div class="flex items-center gap-8">

<div class="text-2xl font-bold tracking-widest text-[#e1fdff] dark:text-cyan-50
font-['Space_Grotesk'] uppercase">

    KINETIC_LAB

</div>

<div class="flex items-center gap-2 px-3 py-1 bg-secondary-container/10 border
border-secondary-container/20">

<div class="w-2 h-2 rounded-full bg-secondary-container animate-pulse"></div>

<span class="font-mono text-[10px] tracking-widest text-secondary-container
uppercase">READY_STATE</span>

```

</div>

</div>

<div class="flex items-center gap-4">

<button class="flex items-center gap-2 px-4 py-1.5 bg-white/5 hover:bg-white/10 transition-all duration-300 group">

sensors

UPLOAD

</button>

<button class="px-6 py-2 bg-primary-container text-on-primary font-['Space_Grotesk'] font-bold tracking-tighter uppercase transition-all duration-300 active:scale-95 brightness-125 hover:shadow-[0_0_20px_rgba(0,242,255,0.3)]">

RUN FILTER

</button>

<div class="flex items-center ml-4 gap-2">

settings

</div>

</div>

</header>

<div class="flex flex-1 overflow-hidden">

<!-- SideNavBar -->

<nav class="fixed left-0 top-0 h-full flex flex-col bg-[#191c22] dark:bg-slate-900 rounded-none border-r border-white/10 w-64 z-40">

<div class="p-8 pb-4">

<div class="flex items-center gap-3 mb-6">

<div class="w-10 h-10 bg-surface-container-highest border border-white/10 flex items-center justify-center overflow-hidden">

</div>

<div>

<div class="font-['Space_Grotesk'] text-xs font-bold uppercase tracking-widest text-primary-container">STATION_01</div>

<div class="font-mono text-[9px] opacity-40">LD_482.990.2</div>

</div>

</div>

<button class="w-full py-3 bg-white/5 hover:bg-white/10 border border-white/5 transition-all text-left px-4 group">

NEW_SESSION

</button>

</div>

<div class="flex-1 mt-4">

analytics

Signal Processing

waves

Noise Analysis

folder_open

Library

terminal

Logs

</div>

<div class="p-6 border-t border-white/5">

<div class="flex items-center justify-between mb-4">

<div class="flex items-center gap-2">

memory

System Health

```
</div>

<span class="font-mono text-[9px] text-secondary-container">98%</span>

</div>

<div class="w-full h-1 bg-white/5">

<div class="h-full bg-secondary-container w-[98%]"></div>

</div>

</div>

</nav>

<!-- Main Content -->

<main class="ml-64 flex-1 flex flex-col bg-surface-container-lowest overflow-y-auto
custom-scrollbar pb-16">

<!-- Contextual Controls -->

<section class="p-6 border-b border-white/5 grid grid-cols-12 gap-6 items-center">

<div class="col-span-3">

<label class="block font-label text-[10px] uppercase tracking-[0.2em] text-on-surface-variant
mb-2">Dataset Selection</label>

<div class="relative group">

<select class="w-full bg-surface-container-low border-0 ghost-border py-2.5 px-4 font-mono
text-xs focus:ring-1 focus:ring-primary-container appearance-none transition-all">

<option>REALTIME_FEED_01.dat</option>

<option>ARCHIVE_BATCH_B.log</option>

<option>SYNTH_WAVEFORM_GEN</option>

</select>

<span class="absolute right-4 top-1/2 -translate-y-1/2 material-symbols-outlined text-sm
opacity-50 pointer-events-none" data-icon="expand_more">expand_more</span>

</div>
```

</div>

<div class="col-span-2">

<label class="block font-label text-[10px] uppercase tracking-[0.2em] text-on-surface-variant mb-2 flex justify-between">

Samples (N) 1024

</label>

<input class="w-full accent-primary-container h-1 bg-surface-container-highest cursor-pointer" type="range"/>

</div>

<div class="col-span-2">

<label class="block font-label text-[10px] uppercase tracking-[0.2em] text-on-surface-variant mb-2 flex justify-between">

Noise Level -12dB

</label>

<input class="w-full accent-primary-container h-1 bg-surface-container-highest cursor-pointer" type="range"/>

</div>

<div class="col-span-3">

<label class="block font-label text-[10px] uppercase tracking-[0.2em] text-on-surface-variant mb-2 flex justify-between">

Filter Order 75 Taps

</label>

<div class="flex items-center gap-4">

<input class="flex-1 accent-primary-container h-1 bg-surface-container-highest cursor-pointer" type="range"/>

<div class="flex items-center gap-2">

AUTO

```
<div class="w-8 h-4 bg-primary-container rounded-full p-0.5 flex justify-end items-center cursor-pointer">
```

```
<div class="w-3 h-3 bg-on-primary-container rounded-full"></div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="col-span-2 flex justify-end">
```

```
<button class="ghost-border px-4 py-2 hover:bg-white/5 transition-all flex items-center gap-2">
```

```
<span class="material-symbols-outlined text-sm" data-icon="tune">tune</span>
```

```
<span class="font-label text-[10px] uppercase font-bold tracking-widest">Advanced</span>
```

```
</button>
```

```
</div>
```

```
</section>
```

```
<!-- Key Metrics -->
```

```
<section class="p-6 grid grid-cols-6 gap-4">
```

```
<div class="bg-surface-container border-l-2 border-primary-container p-4">
```

```
<div class="font-label text-[10px] uppercase tracking-widest text-on-surface-variant mb-4">Filter  
Order</div>
```

```
<div class="font-mono text-2xl font-bold text-primary">75 <span class="text-[10px] opacity-40  
font-normal">taps</span></div>
```

```
</div>
```

```
<div class="bg-surface-container border-l-2 border-primary-container p-4">
```

```
<div class="font-label text-[10px] uppercase tracking-widest text-on-surface-variant mb-4">Step  
Size</div>
```

```
<div class="font-mono text-2xl font-bold text-primary">1.52<span  
class="text-primary-container/60">e-1</span></div>
```

</div>

<div class="bg-surface-container border-l-2 border-error p-4">

<div class="font-label text-[10px] uppercase tracking-widest text-on-surface-variant mb-4">MSE</div>

<div class="font-mono text-2xl font-bold text-[#ff00ff]">2.989e-3</div>

</div>

<div class="bg-surface-container border-l-2 border-primary-container p-4">

<div class="font-label text-[10px] uppercase tracking-widest text-on-surface-variant mb-4">Input SNR</div>

<div class="font-mono text-2xl font-bold text-primary">2.1 dB</div>

</div>

<div class="bg-surface-container border-l-2 border-secondary-container p-4">

<div class="font-label text-[10px] uppercase tracking-widest text-on-surface-variant mb-4">Output SNR</div>

<div class="font-mono text-2xl font-bold text-secondary-container">6.0 dB</div>

</div>

<div class="bg-surface-container border-l-2 border-secondary-container p-4">

<div class="font-label text-[10px] uppercase tracking-widest text-on-surface-variant mb-4">SNR Gain</div>

<div class="font-mono text-2xl font-bold text-secondary-container">+3.9 dB</div>

</div>

</section>

<!-- Main Visualizer -->

<section class="px-6 pb-6 h-[400px]">

```
<div class="h-full bg-surface-container-low ghost-border relative flex flex-col p-6">

<div class="flex justify-between items-start mb-8">

<div>

<h2 class="font-headline text-xl uppercase tracking-tighter mb-1">Adaptive Signal
Reconstruction</h2>

<p class="font-mono text-[10px] opacity-40 uppercase">Domain: Time Frequency / Resolution:
1024-Pt</p>

</div>

<div class="flex gap-6">

<div class="flex items-center gap-2">

<div class="w-3 h-0.5 bg-error"></div>

<span class="font-mono text-[10px] uppercase text-on-surface-variant">Noisy Input</span>

</div>

<div class="flex items-center gap-2">

<div class="w-3 h-0.5 bg-secondary-container"></div>

<span class="font-mono text-[10px] uppercase text-on-surface-variant">LMS Output</span>

</div>

<div class="flex items-center gap-2">

<div class="w-3 h-0.5 bg-primary-container border-b-2 border-dashed
border-primary-container/0"></div>

<span class="font-mono text-[10px] uppercase text-on-surface-variant">Lowpass Est</span>

</div>

</div>

</div>

<div class="flex-1 relative border-l border-b border-white/10 ml-10 mb-10 overflow-hidden">

<!-- Grid Lines -->
```

```
<div class="absolute inset-0 grid grid-cols-12 pointer-events-none opacity-20">
```

```
<div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div><div class="border-r border-white/20"></div>
```

```
</div>
```

```
<div class="absolute inset-0 grid grid-rows-6 pointer-events-none opacity-20">
```

```
<div class="border-b border-white/20"></div><div class="border-b border-white/20"></div><div class="border-b border-white/20"></div><div class="border-b border-white/20"></div><div class="border-b border-white/20"></div><div class="border-b border-white/20"></div>
```

```
</div>
```

```
<!-- Data Visualizations (Simulated SVG) -->
```

```
<svg class="absolute inset-0 w-full h-full" preserveaspectratio="none" viewBox="0 0 1000 400">
```

```
<!-- Lowpass Est (Blue Dashed) -->
```

```
<path d="M0,200 Q100,100 200,200 T400,200 T600,200 T800,200 T1000,200" fill="none" opacity="0.6" stroke="#00f2ff" stroke-dasharray="8,4" stroke-width="1.5"></path>
```

```
<!-- Noisy Input (Red/Pink) -->
```

```
<path d="M0,200 L10,180 L20,230 L30,170 L40,240 L50,160 L60,220 L70,190 L80,210 L90,150 L100,200 L110,160 L120,240 L130,180 L140,220 L150,140 L160,260 L170,190 L180,210 L190,150 L200,200 L210,160 L220,240 L230,180 L240,220 L250,140 L260,260 L270,190 L280,210 L290,150 L300,200 L310,160 L320,240 L330,180 L340,220 L350,140 L360,260 L370,190 L380,210 L390,150 L400,200 L410,160 L420,240 L430,180 L440,220 L450,140 L460,260 L470,190 L480,210 L490,150 L500,200 L510,160 L520,240 L530,180 L540,220 L550,140 L560,260 L570,190 L580,210 L590,150 L600,200 L610,160 L620,240 L630,180 L640,220 L650,140 L660,260 L670,190 L680,210 L690,150 L700,200 L710,160 L720,240 L730,180 L740,220 L750,140 L760,260 L770,190 L780,210 L790,150 L800,200 L810,160 L820,240 L830,180 L840,220 L850,140 L860,260 L870,190 L880,210 L890,150 L900,200 L910,160 L920,240 L930,180 L940,220 L950,140 L960,260 L970,190 L980,210 L990,150 L1000,200" fill="none" opacity="0.4" stroke="#ffb4ab" stroke-width="1"></path>
```

```
<!-- LMS Output (Green) -->
```

```
<path d="M0,200 Q50,180 100,210 T200,190 T300,205 T400,195 T500,200 T600,198
T700,202 T800,200 T900,199 T1000,200" fill="none" stroke="#2ff801" stroke-width="2"></path>
```

```
</svg>
```

```
<!-- Y-Axis Labels -->
```

```
<div class="absolute -left-10 inset-y-0 flex flex-col justify-between text-[8px] font-mono
text-slate-500 py-2">
```

```
<span>+1.0V</span>
```

```
<span>+0.5V</span>
```

```
<span>0.0V</span>
```

```
<span>-0.5V</span>
```

```
<span>-1.0V</span>
```

```
</div>
```

```
<!-- X-Axis Labels -->
```

```
<div class="absolute -bottom-8 inset-x-0 flex justify-between text-[8px] font-mono text-slate-500
px-2">
```

```
<span>0ms</span>
```

```
<span>100ms</span>
```

```
<span>200ms</span>
```

```
<span>300ms</span>
```

```
<span>400ms</span>
```

```
<span>500ms</span>
```

```
</div>
```

```
</div>
```

```
<!-- Floating Metadata -->
```

```
<div class="absolute bottom-16 right-16 bg-surface-container-highest/80 backdrop-blur-md p-4
border border-white/10 max-w-[200px]">
```


<div class="font-label text-[9px] uppercase tracking-widest text-primary-container mb-2 flex items-center gap-2">

info

Analysis Node 7A

</div>

<div class="space-y-2">

<div class="flex justify-between font-mono text-[9px]">

STDEV:

0.0412

</div>

<div class="flex justify-between font-mono text-[9px]">

KURT:

3.125

</div>

<div class="flex justify-between font-mono text-[9px]">

VAR:

0.0017

</div>

</div>

</div>

</div>

</section>

<!-- New Analysis Sections -->

<section class="px-6 grid grid-cols-12 gap-6 mb-8">

<!-- Compound Match Analysis -->

<div class="col-span-7 bg-surface-container-low ghost-border p-6">

<div class="flex items-center justify-between mb-6">

<h3 class="font-headline text-sm font-bold uppercase tracking-widest text-primary-container flex items-center gap-2">

biotech

Compound Match Analysis

</h3>

<div class="font-mono text-[10px] text-slate-500">DATABASE: NIST_MAIN_2023</div>

</div>

<div class="overflow-x-auto">

<table class="w-full text-left font-mono text-xs">

<thead>

<tr class="border-b border-white/5 text-on-surface-variant">

<th class="pb-3 font-medium uppercase tracking-tighter">Identity</th>

<th class="pb-3 font-medium uppercase tracking-tighter">CAS Number</th>

<th class="pb-3 font-medium uppercase tracking-tighter">Match %</th>

<th class="pb-3 font-medium uppercase tracking-tighter">Confidence</th>

</tr>

</thead>

<tbody class="divide-y divide-white/5">

<tr class="group hover:bg-white/5 transition-colors">

<td class="py-4 text-primary font-medium">Ethanol Absolute</td>

<td class="py-4 opacity-40">64-17-5</td>

<td class="py-4 text-secondary-container">98.2%</td>

<td class="py-4">

```
<div class="w-16 h-1 bg-white/5 overflow-hidden">

<div class="h-full bg-secondary-container w-[98%]"></div>

</div>

</td>

</tr>

<tr class="group hover:bg-white/5 transition-colors">

<td class="py-4 text-primary font-medium">Isopropanol</td>

<td class="py-4 opacity-40">67-63-0</td>

<td class="py-4 text-secondary-container/60">14.1%</td>

<td class="py-4">

<div class="w-16 h-1 bg-white/5 overflow-hidden">

<div class="h-full bg-secondary-container w-[14%]"></div>

</div>

</td>

</tr>

<tr class="group hover:bg-white/5 transition-colors">

<td class="py-4 text-primary font-medium">Acetonitrile</td>

<td class="py-4 opacity-40">75-05-8</td>

<td class="py-4 text-secondary-container/20">3.4%</td>

<td class="py-4">

<div class="w-16 h-1 bg-white/5 overflow-hidden">

<div class="h-full bg-secondary-container w-[3%]"></div>

</div>

</td>
```

</tr>

</tbody>

</table>

</div>

</div>

<!-- Functional Groups Identified -->

<div class="col-span-5 bg-surface-container-low ghost-border p-6 flex flex-col">

<h3 class="font-headline text-sm font-bold uppercase tracking-widest text-[#ff00ff] mb-6 flex items-center gap-2">

hub

Functional Groups Identified

</h3>

<div class="flex flex-wrap gap-2">

<div class="px-3 py-1.5 bg-secondary-container/10 border border-secondary-container/30 text-secondary-container font-mono text-[10px] uppercase tracking-wider flex items-center gap-2">

O-H Stretch

</div>

<div class="px-3 py-1.5 bg-primary-container/10 border border-primary-container/30 text-primary-container font-mono text-[10px] uppercase tracking-wider flex items-center gap-2">

C-H Alkanes

</div>

<div class="px-3 py-1.5 bg-white/5 border border-white/10 text-slate-400 font-mono text-[10px] uppercase tracking-wider flex items-center gap-2">

C-O Bend

</div>

<div class="px-3 py-1.5 bg-primary-container/10 border border-primary-container/30 text-primary-container font-mono text-[10px] uppercase tracking-wider flex items-center gap-2">

CH2 Scissoring

</div>

<div class="px-3 py-1.5 bg-white/5 border border-white/10 text-slate-400 font-mono text-[10px] uppercase tracking-wider flex items-center gap-2">

Methyl Group

</div>

</div>

<div class="mt-auto pt-6 border-t border-white/5">

<div class="flex justify-between items-center">

Spectral Integrity

HIGH_RESOLVE

</div>

<div class="mt-2 w-full h-1 bg-white/5">

<div class="h-full bg-[#ff00ff] w-[82%]"></div>

</div>

</div>

</div>

</section>

</main>

</div>

<!-- Footer -->

<footer class="fixed bottom-0 right-0 left-64 flex justify-between px-6 py-2 z-40 bg-[#10131a] dark:bg-black rounded-none border-t border-white/5 items-center">

<div class="font-mono text-[10px] uppercase tracking-tighter text-[#2ff801] dark:text-green-400">

CORE_OS v2.4.0 // SIGNAL_PROCESSING_LAB

</div>

<div class="flex gap-6">

API_DOCS

LATENCY_REPORT

NODE_STATUS

</div>

</footer>

</body></html>